

forth. For evaluation, the ranking of the scale s that a belongs to ($rank_s$) is considered. We consider Mean Reciprocal Rank (MRR) as the evaluation metric; see Equation (1).

$$MRR = \frac{1}{|D_a|} \sum_{s \in D_a} \frac{1}{rank_s} \quad (1)$$

Intuitively, the higher the scale that an adjective belongs to is ranked, the closer MRR will be to 1. For instance, if all adjectives are aligned on their corresponding scales, the MRR will be 1. If models fail to rank the correct scales as closest, it means that they do not encode word polysemes and fine-grained distinction among different scales.

3.2.2. Scale Membership Direct Probing Experiment and Results

When target adjectives are tokenized as multiple segments, token representations are averaged as the adjectives’ final representations. We use layer-wise adjective word embeddings to evaluate information in each layer (e.g. $\vec{a}$ and $\vec{s}$ have 12 representations using a 12-layer model). The experiment is repeated ten times with different random seeds for context sentences. We use fast-text static word embeddings⁵ (Mikolov et al., 2018) trained on 600B Common Crawl data as a baseline. For the baseline, we add the representations of strongest and weakest adjectives on s as $\vec{s}$, then also rank $\cos(\vec{a}, \vec{s})$ for each a to calculate MRR. Here, we only report the results for the best-performing layer for non-baseline models in Table 2.⁶

Models	DM	CD	WK
fast-text	0.842	0.716	0.983
BERT-b	0.829 $\pm$ 0.010	0.797 $\pm$ 0.010	0.997 $\pm$ 0.004
BERT-l	0.853 $\pm$ 0.007	0.805 $\pm$ 0.011	0.997 $\pm$ 0.006
RoBERTa-b	0.668 $\pm$ 0.014	0.705 $\pm$ 0.007	0.906 $\pm$ 0.018
RoBERTa-l	0.777 $\pm$ 0.011	0.757 $\pm$ 0.008	0.977 $\pm$ 0.010

Table 2: Direct scale membership probing results, subscripted numbers are standard deviation in ten runs. The best results per dataset are in bold.

The fast-text baseline is quite strong considering its training data size. All LLMs encode rich scale information as they mostly outperform (with some behaving on par with) the baseline. Model-wise, BERT models encode more information than RoBERTa. Within the same architecture, the bigger a model is, the better they are. Across datasets, **WK** is relatively easy because it (i) has fewer polysemous adjectives with the same forms on different

scales and (ii) has fewer scales for classification than the other two datasets.

3.2.3. Scale Membership Indirect Probing Method

Inspired by Lorge and Pierrehumbert (2023) and de Melo and Bansal (2013), we use four templates including ADJ_{weak} , *if not* ADJ_{strong} to indirectly assess LLMs’ knowledge about scale membership. Adjectives on the same scale are more likely to co-occur in these constructions than those that are not (e.g. *warm*, *if not hot* should be more likely to appear than *warm*, *if not thin*).⁷

For each adjective on a half scale that is not the strongest item, we use it as ADJ_{weak} in our templates and obtain 5 most likely words as ADJ_{strong} . For instance, for scale *adequate-fine-fitting-good*, we prompt models to answer what is likely to be ADJ_{strong} in *adequate/fine/fitting, if not* ADJ_{strong} . We consider a case to be correct if any top-5 word in ADJ_{strong} is on the same scale with ADJ_{weak} , and the overlap is not trivial (e.g. *adequate or even adequate* is a trivial completion, *good or even adequate* is considered to be correct despite the wrong intensity ranking).

3.2.4. Scale Membership Indirect Probing Experiment and Results

For encoder models, we put [MASK] and a comma in the position of ADJ_{strong} (e.g. ADJ_{weak} , *if not* [MASK],) as preliminary experiment results show that BERT mostly predicts punctuation marks as top 5 predictions for [MASK] when it appears as the last token. The comma is used because it does not constrain whether its preceding adjective is attributive (i.e. *can* be immediately followed by a noun) or predicative (i.e. *cannot* be immediately followed by a noun). For non-baseline models except GPT-4, we directly generate ADJ_{strong} given the preceding prompt. We use a similar generation objective for GPT-4 with the best template in other models (see appendix 11.7). We use the Google Ngram corpus slice for 2019 (Lin et al., 2012) as the indirect probing baseline with a similar setting, to obtain the 5 most likely completions for ADJ_{strong} in each construction. In Table 3, we report the best results among all templates in the assessed dataset (we report results based on selecting the best templates with held-out datasets in Table 13).

Indirect probing always underperforms direct probing for all models. This result aligns with Petroni et al. (2019). The relative rankings of different models are largely consistent. BERT is better than RoBERTa given the same size class, and larger encoder models are better than smaller ones

⁵<https://github.com/facebookresearch/fastText/>

⁶We also tried other alternative methods to compute membership, which does not generally work as well as the method we report here. See Appendix 11.1.

⁷See Appendix 11.3 for all templates.

Models	DM	CD	WK
Google Ngram	0.287 ₁	0.075 ₁	0.368 ₁
BERT-b	0.425 ₁	0.066 ₁	0.167 ₁
BERT-l	0.504 ₁	0.109 ₃	0.250 ₁
RoBERTa-b	0.271 ₄	0.064 ₃	0.114 ₁
RoBERTa-l	0.463 ₁	0.100 ₁	0.286 ₁
Falcon	0.265 ₁	0.045 ₁	0.167 ₁
GPT-4	0.540	0.273	0.500
Flan-T5-xl	0.544 ₃	0.177 ₃	0.500 ₃
Flan-T5-xxl	0.515 ₃	0.156 ₃	0.364 ₃

Table 3: Indirect probing results for scale membership. The best results per dataset are in bold. Subscripts refer to the indices of the best templates.

with the same architecture. These observations support the validity of our methods.

We find that all models except Falcon encode rich information about scale membership, in that they either behave on par with the baseline or outperform it. GPT-4 and Flan-T5-xl each achieve the best results on two benchmarks. While GPT-4 has competitive results with Flan-T5-xl in **DM** and **WK**, it is much better for **CD**. Therefore, we conclude that GPT-4 is the best-performing model in this task. Within the same model family, the scaling law tends to hold (i.e., larger is better). However, scaling does not fully explain the pattern of results. For instance, Falcon underperforms much smaller encoder models and even the baseline, and Flan-T5-xxl underperforms Flan-T5-xl.

3.3. Probing Scalar Intensity

In this section, we describe how we probe whether LLMs understand that SAs on the same scale have varied intensities (e.g. *hot* denotes a higher temperature than *warm*).

3.3.1. Scalar Intensity Direct Probing Method

Unlike previous methods which ground SAs in identical contexts and calculate them in separate runs (e.g. It is a *good/great/wonderful/awesome* movie) (Garí Soler and Apidianaki, 2020) (G&A), we use a novel method, which involves binding all SAs on the same scale as one single input to obtain their contextualized representations (e.g. *good great wonderful awesome*), as illustrated in Figure 2. Our assumption is that people are more likely to notice scalar words have different intensities when they are salient in the same context (Ronai and Xiang, 2023a).

For each half-scale s , we shuffle and bind all SAs a on s ten times to get 10 different inputs (e.g. *good great wonderful awesome, great wonderful good awesome*, etc.). For each a , we average the representations of it in 10 inputs to get its final

representation. This is to avoid potential heuristic associations between word order and scalar intensity.

After getting the representation of all a , we utilize the best method (DiffVec) in Garí Soler and Apidianaki (2020) (G&A) to get the intensity ranking for them. Specifically, the intensity of all adjectives a in dataset D_a is measured by their cosine similarities to an external global intensity vector d_{Vec} derived from another SA dataset D_{vec} ($D_{vec} \neq D_a$).⁸

For every half-scale s in D_{vec} , the mildest and the extreme adjectives on s are notated as a_{ms} and a_{es} . We calculate d_{Vec} with Equation (2).

$$d_{Vec} = \frac{1}{|D_{vec} \setminus D_a|} \sum_{s \in D_{vec} \setminus D_a} a_{es} - a_{ms} \quad (2)$$

For all $s \in D_a$, adjectives on s are ranked by their cosine similarities to d_{Vec} : higher similarity means higher intensity. If two adjectives have the same similarity, they are treated as equally intense.

3.3.2. Scalar Intensity Direct Probing Experiment and Results

We probe layer-wise word embeddings in BERT-b/l, RoBERTa-b/l and report the best results for pair-wise accuracy across layers using d_{Vec} in different datasets. We again provide fast-text embeddings as a baseline here. For the baseline, we use static embeddings of a_{es} and a_{ms} for all $s \in D_{vec} \setminus D_a$ to derive d_{Vec} as in Equation 2.

Results are averaged over adjective pairs and shown in Table 4.

Model	Method	DM	CD	WK
fast-text	-	0.637 _{WK}	0.685 _{DM}	0.836 _{CD}
BERT-b	G&A	0.646 _{CD}	0.735 _{DM}	0.902 _{DM}
	Ours	0.639 _{CD}	0.706 _{DM}	0.967_{DM}
BERT-l	G&A	0.695_{CD}	0.731 _{DM}	0.918 _{DM}
	Ours	0.673 _{CD}	0.727 _{DM}	0.902 _{DM}
RoBERTa-b	G&A	0.557 _{WK}	0.645 _{DM}	0.820 _{DM}
	Ours	0.648 _{CD}	0.748 _{DM}	0.934 _{DM}
RoBERTa-l	G&A	0.595 _{CD}	0.682 _{DM}	0.836 _{DM}
	Ours	0.664 _{CD}	0.752_{DM}	0.934 _{DM}

Table 4: Direct scalar intensity probing with our method and G&A on different models. Subscripted letters refer to the source dataset of d_{Vec} . The best results per dataset are in bold. The best results using the same model are underlined.

We find that our simple method is generally better than G&A.⁹ Globally, our method achieves the best-ranking results in two out of three datasets. Although G&A marginally outperforms our method

⁸We don't use the global intensity vector in D_a due to the concern of generalisability.

⁹Our method is also superior to Garí Soler and Apidianaki (2020) in other metrics. See appendix 11.2.

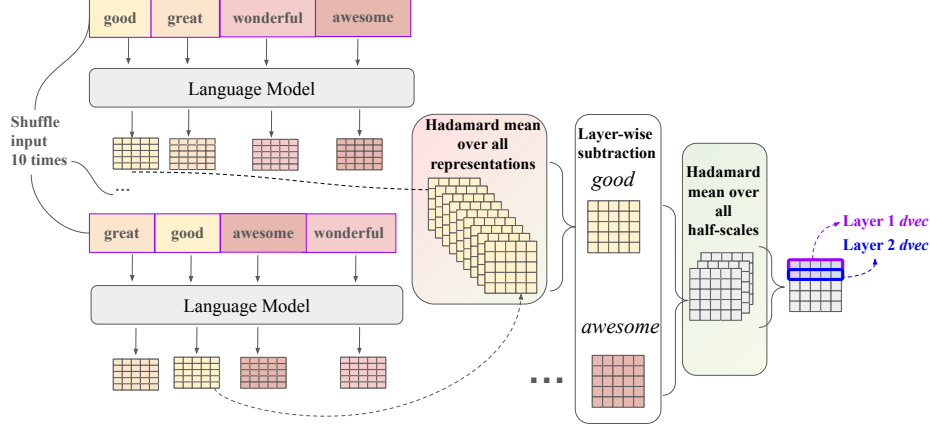


Figure 2: The process to derive intensity vector $\vec{d}_{vec}$. First, an adjective half-scale is randomly shuffled ten times for the order of adjectives as inputs to a language model. Then the encoded word vectors for the same word in different inputs are conducted with the Hadamard mean to derive the final representation of the word. After that, intensity vector $\vec{d}_{vec}$ is calculated by subtracting layer-wise representation of the weakest adjective from the strongest adjective (*awesome* – *good* in this case) then averaging over all relevant half-scale subtractions in a dataset. Then layer-wise $\vec{d}_{vec}$ is used to probe language models’ knowledge for adjective intensities.

in some of the datasets in BERT, our method uniformly largely outperforms G&A in RoBERTa, showing that it is more robust across different models and datasets.

All models encode rich information about SA intensity by outperforming the baseline. Across models, BERT models again dominate in two out of three datasets, showing that they encode more intensity information than RoBERTa, which indicates that larger models are not always better. Across datasets, **WK** is again relatively easy because it does not contain ties.

3.3.3. Scalar Intensity Indirect Probing Method

To complement closed-source models, we indirectly probe intensity using a similar objective as in [Lorge and Pierrehumbert \(2023\)](#). Specifically, for each pair of SAs to compare intensities, we compare the perplexity of minimal-pair prompts containing these adjectives such as ‘*good* but not *awesome*’ and ‘*awesome* but not *good*’, with the former featuring correct order of scalar adjectives and the latter incorrect. Ideally, models should always compute lower perplexity for the first prompt, as the latter is infelicitous or even ungrammatical.

We adopt all 34 templates used in the data collection of [de Melo and Bansal \(2013\)](#).¹⁰ For each

¹⁰Except $ADJ_{strong}(,)$ or *very* ADJ_{weak} because some adjectives examined in this study are extreme ones ([Paradis, 1998](#)) which cannot be modified by *very* (e.g. *very*

template, we compare the perplexity of correct and incorrect constructions. We consider that LLMs encode the scalar intensity for the corresponding adjective pair if the correct one has lower perplexity. We consider two adjectives to be equally intense if switching their positions in a template produces equally unlikely phrases.

3.3.4. Scalar Intensity Indirect Probing Experiment and Results

For encoder models, we estimate pseudo-perplexity by masking out tokens one by one and multiplying the likelihood of the original token appearing in the mask position to compute the sequence’s pseudo-likelihood ([Salazar et al., 2020](#)). For other models except GPT-4, we directly compute perplexity for each sequence. For GPT-4, we use the best-performing template for other models and mimic the setting as in Appendix 11.7. As a baseline, we use Google Ngram corpus slice in 2019 to retrieve the likelihood of constructions as an analogy to perplexity. We then consider a case to be classified correctly if the correct construction is more likely to appear than the incorrect one. We use pair-wise accuracy as an evaluation metric and report the best results among all templates in the assessed dataset. Final results are averaged over pairs and shown in Table 5 (we report results based on selecting the best templates with held-out datasets in Table 14).

terrific). See Appendix 11.4 for the full template list.